Towards Formal Security Proofs of MQOM in the Random Oracle Model plus Ideal-Cipher Model

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Abstract: Recent MPC-in-the-Head (MPCitH) signatures increasingly rely on aggressive GGM-tree optimizations to reduce signature size and cost, culminating in *secret-key-root correlated* GGM (scGGM) tree as used in SBC (Huth and Joux, CRYPTO 2024), MQOM (NIST PQC Standardization for Additional Signature Round-2, 2024), and rBN++ (Kim, Lee, and Son, EUROCRYPT 2025). While this optimization provides substantial compression, it also introduces a dependency loop in existing security reductions, leaving the formal security of scGGM-based constructions challenging (Kosuge and Xagawa, ePrint 2025/1999).
Kosuge and Xagawa (SCIS 2026) established simulation security of scGGM-based all-but-one vector commitments (AVCs) in the (quantum) random-oracle model by heuristically modeling block-cipher-based hash functions as random oracles. However, proving the same simulation-security property in the ideal-cipher model (ICM) remained open.
In this paper, we prove the simulation security of a slightly modified version of scGGM-based AVC used in MQOM in the ICM. For the GF(2) parameter sets, our proof relies on the standard one-wayness of the key-generation problem and a heuristic matrix-rank conjecture. The proof combines the H-coefficient technique with computational one-wayness to bound the probability of bad transcripts, which may be of independent interest. As a consequence, we obtain an EUF-CMA security proof for the modified MQOM in the ROM+ICM.

Keywords: MPC-in-the-Head signatures, ideal cipher model, H-coefficient technique.

1. Introduction

The MPC-in-the-Head (MPCitH) paradigm has become one of the most successful approaches for constructing post-quantum signature schemes [IKOS07][GMO16]. Its importance is illustrated by the fact that several candidates in the NIST additional standardization process are based on MPCitH and its variants [BBFR26][ABB+26][BBB+26]. Recent MPCitH frameworks achieve compact signatures while relying on conservative assumptions such as the hardness of solving multivariate quadratic systems, syndrome decoding problems, or the security of symmetric primitives.

A crucial component of modern MPCitH signatures is the all-but-one vector commitment (AVC) [BBD+23], which is typically instantiated using GGM trees. To reduce signature size and computational cost, several optimizations of GGM trees have been proposed, including the correlated GGM tree [GYW+23]. Among them, the secret-key-root correlated GGM tree (scGGM tree) [HJ24][KLS25] has attracted considerable attention because it reduces the signature size. This optimization has been adopted by MQOM [BBFR24] as well as several recent MPCitH signature schemes [HJ24][KLS25][HJ25].

Despite its practical advantages, establishing the security of scGGM-based constructions has turned out to be surprisingly challenging. Kosuge and Xagawa [KX25] pointed out a gap in the existing EUF-CMA security proofs of MQOM and related schemes. The issue originates from the fact that the root of the correlated tree is derived from the signing key itself. In a standard game-hopping argument, one would like to replace the hidden leaf seed or tape by a random value. However, establishing the validity of this replacement is precisely the hiding property that the proof seeks to establish, leading to a circular argument. Similar difficulties arise not only for MQOM but also for other MPCitH signatures employing correlated-tree techniques.

To address this problem, recent work has advocated proving a simulation property for the AVC component. Simulation security informally states that transcripts generated from genuine scGGM openings and the signing key are computationally indistinguishable from transcripts generated by a simulator that does not know the signing key [KX25]. Once such a simulation theorem is established, it can be used as a modular building block in reductions from EUF-CMA security to weaker security notions such as EUF-NMA. While [KX26b] established simulation security for scGGM-based AVCs by heuristically modeling the block-cipher-based hash function as a random oracle, proving the same property in the more appropriate ICM setting remains

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open.

1.1 Our Contribution

In this paper, we focus on the ideal-cipher-based instantiation of MQOM and study the simulation security of its scGGM-based AVC. To facilitate the analysis, we consider a slightly modified AVC construction and the corresponding modified signature scheme MQOM′. Specifically, we exclude the value 0 from the evaluation domain used in the AVC and incorporate a salt into the hash input used in the signing algorithm. The latter modification was already introduced in [KX26b], whereas the former is specific to the present work.

Our proof employs the H-coefficient technique [Pat09][CS14]. Roughly speaking, we classify transcripts into good and bad ones, show that good transcripts are distributed identically in the real and simulated worlds, and bound the probability of bad transcripts via a reduction to the standard one-wayness of the MQOM key-generation problem together with a conjecture on the rank of certain matrices arising in the analysis. This yields a simulation theorem for the modified scGGM-based AVC in the ideal cipher model.

As a consequence, by combining our simulation theorem with the generic framework of [KX25], we obtain an EUF-CMA security proof for MQOM′ in the ROM+ICM setting for the GF(2) parameter sets. The detailed reduction and complete proofs are provided in the full version [KX26a], which contains the full versions of both [KX26b] and the present paper. The purpose of the present paper is to highlight the simulation-security viewpoint and the ideal-cipher analysis of the underlying AVC.

Concurrent with our work, [FR26] presented the first comprehensive security analysis of MPCitH signatures based on scGGM trees. Their analysis introduces simulation-security notions for scGGM-tree-based AVCs and derives security proofs in the ideal cipher model under assumptions including partial-guessing one-wayness (PGOW) and relation-guessing one-wayness (RGOW). In particular, their analysis applies directly to the original MQOM specification. In contrast, our analysis focuses on the specific AVC construction underlying MQOM′ and establishes simulation security using the H-coefficient technique together with standard one-wayness and a matrix-rank conjecture.

2. Preliminaries

The security parameter is denoted by $\lambda \in \mathbb{Z}^+$. We use the standard O-notations. For $n \in \mathbb{Z}^+$, we let $[n] \coloneqq \{0, \ldots, n-1\}$. For $n_1, n_2 \in \mathbb{Z}^+$, we let $[n_1 : n_2] \coloneqq \{n_1, n_1+1, \ldots, n_2-1\}$. PPT and QPT stand for probabilistic and quantum polynomial time, respectively.

For a distribution D, we often write "$x \leftarrow D$," which indicates that we take a sample x according to D. For a finite set $\mathcal{S}$, $U(\mathcal{S})$ denotes the uniform distribution over $\mathcal{S}$. We often write "$x \leftarrow \mathcal{S}$" instead of "$x \leftarrow U(\mathcal{S})$." If inp is a string, then "$\mathsf{out} \leftarrow \mathsf{A}^O(\mathsf{inp})$" denotes the output of algorithm A running on input inp with access to a set of oracles O. If A and oracles are deterministic, then out is a fixed value and we write "$\mathsf{out} \coloneqq \mathsf{A}^O(\mathsf{inp})$." We also use the notation "$\mathsf{out} \coloneqq \mathsf{A}(\mathsf{inp}; r)$" to make the randomness r of A explicit.

Game $\mathsf{Expt}^{\text{euf-cma}}_{\mathsf{DS},\mathcal{A}}(1^\lambda)$	SIGN(msg)
1 : $\mathcal{Q} \coloneqq \emptyset$	1 : $\sigma \leftarrow \mathsf{Sign}(\mathtt{sk}, \mathtt{msg})$
2 : $(\mathtt{vk}, \mathtt{sk}) \leftarrow \mathsf{Gen}(1^\lambda)$	2 : $\mathcal{Q} \coloneqq \mathcal{Q} \cup \{\mathtt{msg}\}$
3 : $(\mathtt{msg}^*, \sigma^*) \leftarrow \mathcal{A}^{\text{SIGN}}(\mathtt{vk})$	3 : **return** σ
4 : **if** $\mathtt{msg}^* \in \mathcal{Q}$ **then**	
5 : **return** 0	
6 : **return** $\mathsf{Vrfy}(\mathtt{vk}, \mathtt{msg}^*, \sigma^*)$	

Fig. 1 Games for EUF-CMA security of signature scheme (Definition 2.2).

2.1 Digital Signature

The model for digital signature schemes is summarized as follows:

Definition 2.1. *A digital signature scheme* DS *consists of the following triple of PPT algorithms* $(\mathsf{Gen}, \mathsf{Sign}, \mathsf{Vrfy})$*:*

- $\mathsf{Gen}(1^\lambda) \to (\mathtt{vk}, \mathtt{sk})$*: a key-generation algorithm that, on input* 1^λ*, outputs a pair of keys* $(\mathtt{vk}, \mathtt{sk})$*.* $\mathtt{vk}$ *and* $\mathtt{sk}$ *are verification and signing keys, respectively.*
- $\mathsf{Sign}(\mathtt{sk}, \mathtt{msg}) \to \sigma$*: a signing algorithm that takes as input signing key* $\mathtt{sk}$ *and message* $\mathtt{msg} \in \mathcal{M}$ *and outputs signature* $\sigma \in \mathcal{S}$*.*
- $\mathsf{Vrfy}(\mathtt{vk}, \mathtt{msg}, \sigma) \to \mathsf{true}/\mathsf{false}$*: a verification algorithm that takes as input verification key* $\mathtt{vk}$*, message* $\mathtt{msg} \in \mathcal{M}$*, and signature* σ *and outputs its decision* 1 *for acceptance or* 0 *for rejection.*

We require perfect correctness; that is, for any key pair $(\mathtt{vk}, \mathtt{sk}) \leftarrow \mathsf{Gen}(1^\lambda)$*, for any message* $\mathtt{msg} \in \mathcal{M}$*, and for any signature* $\sigma \leftarrow \mathsf{Sign}(\mathtt{sk}, \mathtt{msg})$*,* $\mathsf{Vrfy}(\mathtt{vk}, \mathtt{msg}, \sigma) = 1$ *holds.*

Security notions: We review the standard security notion, existential unforgeability against chosen-message attack (EUF-CMA). We also consider a weak version, existential unforgeability against no-message attack (EUF-NMA), in which the adversary cannot access the signing oracle. The formal definition follows:

Definition 2.2 (EUF-NMA and EUF-CMA security)**.** *Let* $\mathsf{DS} = (\mathsf{Gen}, \mathsf{Sign}, \mathsf{Vrfy})$ *be a digital signature scheme. For any* $\mathcal{A}$*, we define its* EUF-CMA *advantage against* DS *as*

$$\mathsf{Adv}^{\text{euf-cma}}_{\mathsf{DS},\mathcal{A}}(\lambda) \coloneqq \Pr[\mathsf{Expt}^{\text{euf-cma}}_{\mathsf{DS},\mathcal{A}}(1^\lambda) = 1],$$

where $\mathsf{Expt}^{\text{euf-cma}}_{\mathsf{DS},\mathcal{A}}(1^\lambda)$ *is an experiment described in Figure 1. We say that* DS *is* EUF-CMA*-secure if* $\mathsf{Adv}^{\text{euf-cma}}_{\mathsf{DS},\mathcal{A}}(\lambda)$ *is negligible for any QPT adversary* $\mathcal{A}$*.*

For any $\mathcal{A}$*, we define its* EUF-NMA *advantage against* DS *as* $\mathsf{Adv}^{\text{euf-nma}}_{\mathsf{DS},\mathcal{A}}(\lambda) \coloneqq \Pr[\mathsf{Expt}^{\text{euf-nma}}_{\mathsf{DS},\mathcal{A}}(1^\lambda) = 1]$*, where* $\mathsf{Expt}^{\text{euf-nma}}_{\mathsf{DS},\mathcal{A}}(1^\lambda)$ *is the game* $\mathsf{Expt}^{\text{euf-cma}}_{\mathsf{DS},\mathcal{A}}(1^\lambda)$ *without the signing oracle* SIGN*. We say that* DS *is* EUF-NMA*-secure if* $\mathsf{Adv}^{\text{euf-nma}}_{\mathsf{DS},\mathcal{A}}(\lambda)$ *is negligible for any QPT adversary* $\mathcal{A}$*.*

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3. MQOM

We review MQOM (ver.2.1) [BBFR24]. The parameter sets are summarized in Table A·1. Its signing algorithm is depicted in Figure 2. The signing algorithm uses an all-but-one vector commitment (AVC) scheme defined as Figure 3. We omit the key-generation and verification algorithms. We also consider a slightly modified variant MQOM′, which excludes 0 from the evaluation domain and includes the salt in the input to Hash_4; all other parts are identical to those of MQOM. Similarly to Mirath [AAB+24] and RYDE [ABB+24], MQOM employs the Threshold-computation-in-the-Head (TCitH) technique [FR25]. MQOM adopts the half-tree technique [GYW+23], the correlated-tree technique and a further optimization within the GGM tree of AVCs in [HJ24], [KLS25], and [HJ25].

MQOM's signing key is $x \leftarrow \mathbb{F}^n$. The verification key consists of m degree-2 polynomials $F = (f_0, \dots, f_{m-1}) \in (\mathbb{F}[x_0, \dots, x_{n-1}])^m$ with $f_i \colon x \mapsto x^\top A_i x + b_i^\top x - y_i$ and the signer shows that it knows x satisfying $F(x) = 0$.

The underlying ID protocol is designed to show the knowledge of x satisfying $F(x) = 0$. Let $\mathbb{K}$ be the μ-dimensional extension of $\mathbb{F}$ with basis $\beta_0, \dots, \beta_{\mu-1}$. Let $\phi \colon (e_0, \dots, e_{\mu-1}) \in \mathbb{F} \mapsto \sum_{i\in[\mu]} e_i \cdot \beta_i \in \mathbb{K}$. Let $\hat{m} \coloneqq m/\mu \in \mathbb{N}$. We expand ϕ in a natural way. We then define

$$\hat{A}_0 = \phi(A_0, \dots, A_{\mu-1}), \dots, \hat{A}_{\hat{m}-1} = \phi(A_{m-\mu}, \dots, A_{m-1})$$
$$\hat{b}_0 = \phi(b_0, \dots, b_{\mu-1}), \dots, \hat{b}_{\hat{m}-1} = \phi(b_{m-\mu}, \dots, b_{m-1})$$
$$\hat{y}_0 = \phi(y_0, \dots, y_{\mu-1}), \dots, \hat{y}_{\hat{m}-1} = \phi(y_{m-\mu}, \dots, y_{m-1}).$$

From ver. 2.1, MQOM's signer shows the relation $\hat{F}(x) = (0, \dots, 0) \in \mathbb{K}^{\hat{m}}$, where $\hat{F} = (f_0, \dots, f_{\hat{m}-1})$ and

$$\hat{f}_i \colon x \mapsto x^\top \hat{A}_i x + \hat{b}_i^\top x - \hat{y}_i.$$

Due to linearity of ϕ, $F(x) = 0$ is equivalent to $\hat{F}(x) = 0$.

For ease of notation, we let $\Omega = \{\omega_0, \dots, \omega_{N-1}\} \subset \mathbb{K}$ be an evaluation domain, which is genearted by the Gray code in $\mathbb{K}$ with $\omega_0 = 0$.

4. Simulatability of AVC

Following [KX25] and [FR26], we define simulatability of AVC in the context of MQOM as in Figure 4. In Figure 4, the real oracle first computes $\mathtt{path}[e][j]$ and $\mathtt{com}[e][i^*]$, computes the other information by using MQOM.Recon, and then computes $\Delta_x^{(1)}[e]$ and α'_0 appropriately, instead of the original computation in the signing algorithm. We note that this change of computation does not change the distribution. We also note that $\mathtt{node}[\mathtt{path}_{j-1}(i^*)] = S_{j-1} \oplus \delta$, where $S_{j-1} \coloneqq \bigoplus_{j'\in[1:j]} \mathtt{path}[j']$. The ideal oracle randomly chooses $\mathtt{path}[e][j]$ and $\mathtt{com}[e][i^*]$, computes the other information by using MQOM.Recon, and chooses $\Delta_x^{(1)}[e]$ and α'_0 uniformly at random.

Definition 4.1 (AVC simulation advantage of MQOM)**.** *For an adversary $\mathcal{A}$, we define its AVC simulation advantage against* MQOM *as*

$$\mathsf{Adv}^{\mathrm{cavc}}_{\mathsf{MQOM},\mathcal{A}}(\lambda) \coloneqq \left| \begin{array}{l} \Pr[\mathsf{Expt}^{\mathrm{cavc},0}_{\mathsf{MQOM},\mathcal{A}}(\lambda) = 1] \\ \quad - \Pr[\mathsf{Expt}^{\mathrm{cavc},1}_{\mathsf{MQOM},\mathcal{A}}(\lambda) = 1] \end{array} \right|,$$

where the experiments are described in Figure 4.

Conversions and parameter settings:

We explicitly consider linear conversions $\mathsf{B2F} \colon \{0,1\}^k \to \mathbb{F}$, $\mathsf{B2K} \colon \{0,1\}^{k\mu} \to \mathbb{K}$, and $\mathsf{F2K} \colon \mathbb{F} \to \mathbb{K}$, which are linear isomorphisms by our assumption and adopted from those in Section 2.3 of [BBFR25]. We denote those linear inversions as F2B, K2B, and K2F. Due to linearity, we have $\mathsf{F2B}(x_0) \oplus \mathsf{F2B}(x_1) = \mathsf{F2B}(x_0 + x_1)$. We extend their domain and range as vector and matrix spaces: For example, if $x \in \mathbb{F}^n$, we have $\mathsf{F2B}(x) \in \{0,1\}^{kn}$.

MQOM employs a linear orthomorphism σ in their $\mathsf{H}_{\mathsf{avc}} = \mathsf{EncFF}$, which is defined as $\sigma(s_l \,\|\, s_r) \coloneqq (s_l \oplus s_r) \,\|\, s_l$ and $\bar{\sigma}(s_l \,\|\, s_r) = ((s_l \oplus s_r) \,\|\, s_l) \oplus (s_l \,\|\, s_r) = s_r \,\|\, (s_l \oplus s_r)$, where $s_l, s_r \in \{0,1\}^{\lambda/2}$. We note that they are efficiently invertible.

We consider GF(2) parameter sets of MQOM because of the following reasons: Let $\mathbb{F} = \mathrm{GF}(2^k)$ and $\mathbb{K} = \mathrm{GF}(2^{k\mu})$. In all parameter sets, we have $m = n$ and $\hat{m} = m/\mu = n/\mu$. By the way, we have $|x| - \lambda = kn - \lambda = \lambda/4$ for all GF(2) parameters, but it is at least $(3/4)\lambda$ for all parameters. Since our reduction requires $2^{|x|-\lambda}$ is small, we only consider GF(2) parameter sets. If we adopt the RGOW assumption [FR26], there is no problem and reduction does not require $2^{|x|-\lambda}$ factor. Since we only consider GF(2) parameter sets, we have $|\Delta_x^{(1)}[e]| + |\alpha_0[e]| < 2\lambda$ and $n_b = 2$.

We recall that, Line 12 of MQOM.Sign, "$(\bar{x}[e][i], \bar{u}[e][i]) \coloneqq \mathsf{Parse}(\mathtt{seed}[e][i], \mathtt{tape}[e][i])$". The bit length of the LHS is $kn + k\mu\eta$, while $|\mathtt{seed}[e][i]| + |\mathtt{tape}[e][i]| = (1 + n_b)\lambda$, where $kn + k\mu\eta < (1 + n_b)\lambda$ for all parameter sets. For technical reasons, we do not want to truncate $\mathtt{tape}[e][i]$ as in Figure 2. Hence, we consider $\bar{u}'[e][i] \in \mathbb{K}^{\eta'}$ with $\eta' \coloneqq ((1 + n_b)\lambda - kn)/(k\mu)$ and let

$$(\bar{x}[e][i], \bar{u}'[e][i]) \coloneqq \mathsf{Parse}(\mathtt{seed}[e][i], \mathtt{tape}[e][i]).$$

We expand the length of $\alpha_0[e]$ also and we will use $\Gamma' = [\Gamma; O_{(\eta'-\eta)\times\hat{m}}] \in \mathbb{K}^{\eta'\times\hat{m}}$.

The original MQOM uses $\omega_0 = 0$ for both $N = 256$ and $N = 2048$. We will need that every ω_i is a unit and, thus, exclude 0.

- For the parameter sets with $N = 256$ and $\mathbb{K} = \mathrm{GF}(2^8 = 256)$, we remove 0 and use $\Omega' = (\omega'_0, \dots, \omega'_{254}) \coloneqq (\omega_1, \dots, \omega_{255})$. Thus, the evaluation domain size becomes $N = 255$.
- For the parameter sets with $N = 2048$ and $\mathbb{K} = \mathrm{GF}(2^{16})$, we remove 0 and use $\Omega' = (\omega'_0, \dots, \omega'_{2047}) \coloneqq (\omega_1, \dots, \omega_{2048})$.

We finally note that the implementation of E in MQOM [BBFR25].

- For the L1 parameter sets, E is instantiated by the AES-128 encryption algorithm.
- For the L3 parameter sets, E is defined as follows: $E \colon (k, p) \in \{0,1\}^{192} \times \{0,1\}^{192} \mapsto$

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MQOM.Sign($\mathtt{sk}, \mathtt{msg}$)

$(\mathtt{vk}, x) \coloneqq \mathsf{Parse}(\mathtt{sk}); (\mathtt{mseed}_{\mathsf{eq}}, \hat{y}) \coloneqq \mathsf{Parse}(\mathtt{vk})$
$(\{\hat{A}_i\}, \{\hat{b}_i\}) \coloneqq \mathsf{ExpandEquations}(\mathtt{mseed}_{\mathsf{eq}})$
$\mathtt{mseed} \leftarrow \{0,1\}^\lambda; \mathtt{salt} \leftarrow \{0,1\}^\lambda$
$h_{\mathtt{msg}} \coloneqq \mathsf{Hash}_2(\mathtt{msg})$
$(\mathtt{rseed}[0], \ldots, \mathtt{rseed}[\tau-1]) \coloneqq \mathsf{PRG}_{\mathsf{seed}}(\mathtt{mseed})$
$\delta \coloneqq \mathsf{FirstBits}_\lambda(x)$
for $e \in [\tau]$ **do**
  $(\mathtt{com}[e], \mathtt{decom}[e], (m_i)_{i\in[N]}) \coloneqq \mathsf{MQOM.Commit}(\mathtt{salt}, \mathtt{rseed}[e], \delta, e)$
  $h_{\mathtt{com}}[e] \coloneqq \mathsf{Hash}_6(\mathtt{com}[e])$
  // $\bar{x}[e][i] \in \mathbb{F}^n, \bar{u}[e][i] \in \mathbb{K}^\eta$
  for $i \in [N]$ **do**
    $(\bar{x}[e][i], \bar{u}[e][i]) \coloneqq \mathsf{Parse}(m_i)$
  // Compute $P_u(X) = u_0[e] + u_1[e] \cdot X \in (\mathbb{K}[X])^\eta$
  $u_0[e] \coloneqq -\sum_{i\in[N]} \omega_i \cdot \bar{u}[e][i]$ // $u_0[e] \in \mathbb{K}^\eta$
  $u_1[e] \coloneqq \sum_{i\in[N]} \bar{u}[e][i]$ // $u_1[e] \in \mathbb{K}^\eta$
  // Compute $P_x(X) = x_0[e] + x \cdot X \in (\mathbb{K}[X])^n$
  $x_0[e] \coloneqq -\sum_{i\in[N]} \omega_i \cdot \bar{x}[e][i]$ // $x_0[e] \in \mathbb{K}^n$
  $\Delta_x[e] \coloneqq x - \sum_{i\in[N]} \bar{x}[e][i]$
  $\Delta_x^{(1)}[e] \coloneqq (\Delta_x[e])[\lambda:]$
$\mathtt{com}_1 \coloneqq \mathsf{Hash}_7(h_{\mathtt{com}}, \Delta_x^{(1)})$

(continued)

$\Gamma \coloneqq \boldsymbol{I}_\eta \in \mathbb{K}^{\eta\times\eta}$ // 3-round ver. $\eta = \hat{m}$
$\Gamma \coloneqq \mathsf{XOF}_8(\mathtt{com}_1) \in \mathbb{K}^{\eta\times\hat{m}}$ // 5-round ver. $\eta < \hat{m}$
for $e \in [\tau]$ **do**
  for $i \in [\hat{m}]$ **do**
    // Compute $P_t(X) = t_0 + t_1 \cdot X \in (\mathbb{K}[X])^n$
    $t_0 \coloneqq \hat{A}_i \cdot x_0[e]$
    $t_1 \coloneqq \hat{A}_i \cdot x + \hat{b}_i$
    // Compute $P_{z,i}(X) = z_{0,i} + z_{1,i} \cdot X \in \mathbb{K}[X]$
    $z_{0,i} \coloneqq t_0^\top \cdot x_0[e]$
    $z_{1,i} \coloneqq t_0^\top \cdot x + t_1^\top \cdot x_0[e]$
  $z_0 \coloneqq (z_{0,0}, \ldots, z_{0,\hat{m}-1})$
  $z_1 \coloneqq (z_{1,0}, \ldots, z_{1,\hat{m}-1})$
  $\alpha_0[e] \coloneqq u_0[e] + \Gamma \cdot z_0$ // $\alpha_0[e] \in \mathbb{K}^\eta$
  $\alpha_1[e] \coloneqq u_1[e] + \Gamma \cdot z_1$ // $\alpha_1[e] \in \mathbb{K}^\eta$
$\mathtt{com}_2 \coloneqq \mathsf{Hash}_3(\alpha_0, \alpha_1)$
$h_2 \coloneqq \mathsf{Hash}_4(\mathtt{vk}, \mathtt{com}_1, \mathtt{com}_2, h_{\mathtt{msg}})$
// Grinding
$\mathtt{ctr} \coloneqq 0$
$(\mathbf{i}^*, \mathtt{val}) \coloneqq \mathsf{XOF}_5(h_2, \mathtt{ctr})$
while $\mathtt{val} \neq 0$ **do**
  $\mathtt{ctr} \coloneqq \mathtt{ctr} + 1$
  $(\mathbf{i}^*, \mathtt{val}) \coloneqq \mathsf{XOF}_5(h_2, \mathtt{ctr})$
for $e \in [\tau]$ **do**
  $\mathtt{pdecom}[e] \coloneqq \mathsf{cAVC.Open}(\mathtt{decom}, e, \mathbf{i}^*[e])$
$\pi \coloneqq (\mathtt{pdecom}, \Delta_x^{(1)})$
return $\sigma \coloneqq \mathsf{Serialize}(\mathtt{salt}, \mathtt{com}_1, \mathtt{com}_2, \alpha_1, \pi, \mathtt{ctr})$

Fig. 2 Expanded signing algorithm of MQOM (ver.2.1)

MQOM.Commit($\mathtt{salt}, \mathtt{rseed}, \delta, e$)

$(\mathtt{node}[2], \mathtt{node}[3]) \coloneqq (\mathtt{rseed}, \mathtt{rseed} \oplus \delta)$
for $j \in [1:d]$ **do**
  $t_{\mathtt{salt},2,e,j} \coloneqq \mathsf{TweakSalt}(\mathtt{salt}, 2, e, j)$
  for $k \in [2^j : 2^{j+1}]$ **do**
    $\mathtt{node}[2k] \coloneqq \mathsf{H}_{\mathsf{avc}}(\mathtt{node}[k], t_{\mathtt{salt},2,e,j})$
    $\mathtt{node}[2k+1] \coloneqq \mathtt{node}[2k] \oplus \mathtt{node}[k]$
$t_{\mathtt{salt},0,e,0} \coloneqq \mathsf{TweakSalt}(\mathtt{salt}, 0, e, 0)$
$t_{\mathtt{salt},1,e,0} \coloneqq \mathsf{TweakSalt}(\mathtt{salt}, 1, e, 0)$
for $i \in [N]$ **do**
  $\mathtt{seed}[i] \coloneqq \mathtt{node}[N+i]$
  $c_0 \coloneqq \mathsf{H}_{\mathsf{avc}}(\mathtt{seed}[i], t_{\mathtt{salt},0,e,0})$
  $c_1 \coloneqq \mathsf{H}_{\mathsf{avc}}(\mathtt{seed}[i], t_{\mathtt{salt},1,e,0})$
  $\mathtt{com}[i] \coloneqq c_0 \,\|\, c_1$
  $\mathtt{tape}[i] \coloneqq \varepsilon$ // Empty string
  for $k \in [n_b]$ **do**
    $t_{\mathtt{salt},3,e,k} \coloneqq \mathsf{TweakSalt}(\mathtt{salt}, 3, e, k)$
    $\mathtt{tape}[i] \coloneqq \mathtt{tape}[i] \,\|\, \mathsf{H}_{\mathsf{avc}}(\mathtt{seed}[i], t_{\mathtt{salt},3,e,k})$
$\mathtt{decom} \coloneqq (\mathtt{node}, \mathtt{com})$
return $\mathtt{com}, \mathtt{decom}, \mathtt{seed}, \mathtt{tape}$

$\mathsf{H}_{\mathsf{avc}}(s,t) = \mathsf{EncFF}(s,t)$

return $E(\mathtt{key}=t, \mathtt{pt}=s) \oplus \sigma(s)$

$\mathsf{TweakSalt}(\mathtt{salt}, k, e, j)$

$\mathsf{tweak} \coloneqq k + 4e + 256j$
$t_{\mathtt{salt},k,e,j} \coloneqq \mathtt{salt} \oplus \mathsf{bin}_\lambda(\mathsf{tweak})$
return $t_{\mathtt{salt},k,e,j}$

MQOM.Open($\mathtt{decom}, e, i^*$)

Parse $\mathtt{decom} = (\mathtt{node}, \mathtt{com})$
for $j \in [1:d+1]$ **do**
  $\mathtt{path}[j] \coloneqq \mathtt{node}[\mathsf{copath}_j(i^*)]$
return $\mathtt{pdecom} \coloneqq (\mathtt{path}, \mathtt{com}[i^*])$

MQOM.Recon($\mathtt{salt}, \mathtt{pdecom}, e, i^*$)

Parse $\mathtt{pdecom} = (\mathtt{path}, \mathtt{com}[i^*])$
for $j \in [1:d+1]$ **do**
  $\mathtt{node}[\mathsf{copath}_j(i^*)] \coloneqq \mathtt{path}[j]$
for $j \in [1:d]$ **do**
  $t_{\mathtt{salt},2,e,j} \coloneqq \mathsf{TweakSalt}(\mathtt{salt}, 2, e, j)$
  for $k \in [2^j : 2^{j+1}]$ **do**
    if $\mathtt{node}[k] \neq \perp$ **then**
      $\mathtt{node}[2k] \coloneqq \mathsf{H}_{\mathsf{avc}}(\mathtt{node}[k], t_{\mathtt{salt},2,e,j})$
      $\mathtt{node}[2k+1] \coloneqq \mathtt{node}[2k] \oplus \mathtt{node}[k]$
$t_{\mathtt{salt},0,e,0} \coloneqq \mathsf{TweakSalt}(\mathtt{salt}, 0, e, 0)$
$t_{\mathtt{salt},1,e,0} \coloneqq \mathsf{TweakSalt}(\mathtt{salt}, 1, e, 0)$
for $i \in [N] \setminus \{i^*\}$ **do**
  $\mathtt{seed}[i] \coloneqq \mathtt{node}[N+i]$
  $c_0 \coloneqq \mathsf{H}_{\mathsf{avc}}(\mathtt{seed}[i], t_{\mathtt{salt},0,e,0})$
  $c_1 \coloneqq \mathsf{H}_{\mathsf{avc}}(\mathtt{seed}[i], t_{\mathtt{salt},1,e,0})$
  $\mathtt{com}[i] \coloneqq c_0 \,\|\, c_1$
  $\mathtt{tape}[i] \coloneqq \varepsilon$ // Empty string
  for $k \in [n_b]$ **do**
    $t_{\mathtt{salt},3,e,k} \coloneqq \mathsf{TweakSalt}(\mathtt{salt}, 3, e, k)$
    $\mathtt{tape}[i] \coloneqq \mathtt{tape}[i] \,\|\, \mathsf{H}_{\mathsf{avc}}(\mathtt{seed}[i], t_{\mathtt{salt},3,e,k})$
return $\mathtt{com}, (\mathtt{seed}[i])_{i\neq i^*}, (\mathtt{tape}[i])_{i\neq i^*}$

Fig. 3 The scGGM-based AVC instantiation used in MQOM, where $E\colon \{0,1\}^\lambda \times \{0,1\}^\lambda \to \{0,1\}^\lambda$ is a block cipher and $\mathsf{copath}_j(i^*) \coloneqq \lfloor (N+i^*)/2^{d-j} \rfloor \oplus 1$, which is the index of the sibling of the hidden-path node at layer $j \in \{1,\ldots,d\}$.

$\mathsf{Enc}(k \,\|\, 0^{64}, p \,\|\, 0^{64})[0 : 192]$, where Enc is the encryption algorithm of Rijndael-256-256.

- For the L5 parameter sets, E is instantiated by the Rijndael-256-256 encryption algorithm.

Thus, formally speaking, the function E used for the L3 parameter sets is not itself an ideal cipher on $\{0,1\}^{192}$, since

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it is obtained by truncating the output of a 256-bit block cipher. Nevertheless, this causes no issue in our proof. For the L3 parameter sets, the ideal-cipher abstraction is applied to the underlying Rijndael-256-256 encryption algorithm Enc, and all calls to E are interpreted as calls to Enc on inputs padded with 0^{64}, followed by truncation to 192 bits.

5. Security of a modified scGGM-based AVC of MQOM in the ICM

This section proves the simulation security of the slighly modified version of the scGGM-based AVC used in MQOM in the ICM. We first recall the H-coefficient technique. We then discuss the parameter settings and give the proof of our main theorem (Theorem 5.2).

H-coefficient technique: We employ the H-coefficient technique [Pat09][CS14][JN18] to upper-bound the distinguisher's advantage between the real and ideal experiments with oracle sets O_{real} and O_{ideal}. An interaction transcript, denoted by θ, consists of the ordered list of query–answer pairs. We consider deterministic distinguishers. Let Θ_{real} (resp. Θ_{ideal}) denote the transcript random variable in the real (resp. ideal) world. A transcript θ is *attainable* if $\Pr[\Theta_{\mathrm{ideal}} = \theta] > 0$.

Theorem 5.1 (H-coefficient Technique)**.** *We consider the two sets of oracles O_{real} and O_{ideal}. Let $\mathcal{D}$ be a deterministic distinguisher. Let Ω be a set of all attainable transcripts made by the distinguisher $\mathcal{D}$. We then consider a partition Bad and Good of Ω, that is, $\Omega = \mathsf{Bad} \sqcup \mathsf{Good}$. Suppose that there exist non-negative ϵ_{good} and ϵ_{bad} such that for all $\theta \in \mathsf{Good}$, it holds that $\frac{\Pr[\Theta_{\mathrm{real}}=\theta]}{\Pr[\Theta_{\mathrm{ideal}}=\theta]} \geq 1 - \epsilon_{\mathrm{good}}$ and $\Pr[\Theta_{\mathrm{ideal}} \in \mathsf{Bad}] \leq \epsilon_{\mathrm{bad}}$. Then, we have*

$$|\Pr[\mathcal{D}^{O_{\mathrm{real}}} = 1] - \Pr[\mathcal{D}^{O_{\mathrm{ideal}}} = 1]| \leq \epsilon_{\mathrm{good}} + \epsilon_{\mathrm{bad}}.$$

5.1 Main Theorem

We want to show the computational indistinguishability between the two games in Figure 4.

Theorem 5.2. *Let $\mathcal{A}$ be a PPT distinguisher that makes at most Q_E queries to $E^{\pm}$. Suppose that the conjecture (Definition 5.3) holds. There exists an adversary $\mathcal{A}_{\mathrm{ow}}$ satisfying*

$$\mathsf{Adv}^{\mathrm{cavc}}_{\mathsf{MQOM},\mathcal{A}}(\lambda) \leq \mathsf{Adv}^{\mathrm{ow}}_{\mathsf{F}_{\mathsf{MQOM}},\mathcal{A}_{\mathrm{ow}}}(\lambda).$$

The running time of $\mathcal{A}_{\mathrm{ow}}$ is that of $\mathcal{A}$ plus $2^{kn-\lambda} \cdot \mathsf{poly}(Q', \lambda)$, where $Q' \coloneqq Q_E + Q \cdot \tau \cdot ((9/2)N - 4)$.

5.1.1 Proof of Theorem 5.2

We first introduce notation for query-answer transcripts. We maintain the following query-answer lists:

- $\mathcal{Q}_{\mathrm{node}}$ stores information related to $\mathtt{path}[e][j]$. For each opening transcript, and for all $e \in [\tau]$ and $j = 2, \ldots, d$, we add the tuple $(t_{\mathtt{salt},2,e,j-1}, b, S_{j-1}, \mathtt{path}[e][j])$ to $\mathcal{Q}_{\mathrm{node}}$.
- $\mathcal{Q}_{\mathrm{com}}$ stores information related to $\mathtt{com}[e][i^*]$. For each query, and for all $e \in [\tau]$, we add the tuples $(t_{\mathtt{salt},0,e,0}, S_d, c_0)$ and $(t_{\mathtt{salt},1,e,0}, S_d, c_1)$ to $\mathcal{Q}_{\mathrm{com}}$.
- $\mathcal{Q}_{\mathrm{beta}}$ stores information related to $\Delta^{(1)}_x[e]$ and $\alpha'_0[e]$. For each query, and for all $e \in [\tau]$, we add the tuple $(t_{\mathtt{salt},3,e,0}, t_{\mathtt{salt},3,e,1}, S_d, \Delta^{(1)}_x[e], \alpha'_0[e])$ to $\mathcal{Q}_{\mathrm{beta}}$.
- $\mathcal{Q}_E$ stores all queries to the ideal cipher E. Specifically, it contains tuples (t, s, y) such that $E^+(t, s) = y$ or $E^-(t, y) = s$. We note that the computations in $\mathsf{MQOM.Recon}$ in Figure 4 are recorded in $\mathcal{Q}_E$, rather than in $\mathcal{Q}_{\mathrm{node}}$, $\mathcal{Q}_{\mathrm{com}}$, or $\mathcal{Q}_{\mathrm{beta}}$. For each tree, $\mathsf{MQOM.Recon}$ makes $N/2$ queries for nodes, $2(N-1)$ queries for commitments, and $2(N-1)$ queries for tapes to E^+.

The transcript θ for the H-coefficient technique consists of $(\mathtt{vk}, \mathtt{sk})$ together with $\mathcal{Q}_{\mathrm{node}}$, $\mathcal{Q}_{\mathrm{com}}$, $\mathcal{Q}_{\mathrm{beta}}$, and $\mathcal{Q}_E$. If there is any inconsistency among these query-answer pairs, an adversary may distinguish the games. We therefore define *bad transcripts* to capture such events.

Bad transcripts with respect to nodes and commitments:

We define bad transcripts capturing when queries to the ideal cipher E collide with those used to compute tree nodes or commitments. Intuitively, such events occur when an adversary (or $\mathsf{MQOM.Recon}$) queries either the input or the output of E corresponding to these computations.

We consider two types of bad events. The first corresponds to queries to E^+ (forward direction). For example, in the case of commitments, this occurs if there exist $(t, s, *) \in \mathcal{Q}_{\mathrm{com}}$ and $(t, s \oplus \delta, *) \in \mathcal{Q}_E$. The second corresponds to queries to E^- (backward direction). For example, in the case of commitments, this occurs if there exist $(t, s, z) \in \mathcal{Q}_{\mathrm{com}}$ and $(t, *, \sigma(s \oplus \delta) \oplus z) \in \mathcal{Q}_E$.

Formally, the bad transcripts with respect to nodes and commitments are defined as follows:

Definition 5.1. *We define the following four bad events:*

- $\mathsf{Bad}_{\mathrm{node},F}$*: There exist $(t, b, s, z) \in \mathcal{Q}_{\mathrm{node}}$ and $(t, s \oplus \delta, *) \in \mathcal{Q}_E$.*
- $\mathsf{Bad}_{\mathrm{node},B}$*: There exist $(t, b, s, z) \in \mathcal{Q}_{\mathrm{node}}$ and $(t, *, \sigma(s \oplus \delta) \oplus b \cdot (s \oplus \delta) \oplus z) \in \mathcal{Q}_E$.*
- $\mathsf{Bad}_{\mathrm{com},F}$*: There exist $(t, s, z) \in \mathcal{Q}_{\mathrm{com}}$ and $(t, s \oplus \delta, *) \in \mathcal{Q}_E$.*
- $\mathsf{Bad}_{\mathrm{com},B}$*: There exist $(t, s, z) \in \mathcal{Q}_{\mathrm{com}}$ and $(t, *, \sigma(s \oplus \delta) \oplus z) \in \mathcal{Q}_E$.*

Bad transcripts with respect to $\Delta^{(1)}_x$ and α'_0 in $\mathcal{Q}_{\mathrm{beta}}$: Intuitively, we define bad transcripts capturing when queries to the ideal cipher E collide with those used to compute $\Delta^{(1)}_x$ and α'_0. As before, we distinguish between forward (E^+) and backward (E^-) queries.

The first type of bad event corresponds to queries to E^+ (forward direction). This occurs if there exist $(t_0, t_1, s, *, *) \in \mathcal{Q}_{\mathrm{beta}}$ and $(t_b, s \oplus \delta, *) \in \mathcal{Q}_E$ for some $b \in \{0, 1\}$.

The second type of bad event corresponds to queries to E^- (backward direction). If we consider the real world, this captures queries to outputs of the form $\xi_b \oplus \sigma(S_d \oplus \delta)$ for $b \in \{0, 1\}$. Hereafter, we explain how to formalize this second event in the ideal world.

Fix q and e. Let $i^* \coloneqq \mathbf{i}^*[e]$, and define $t_0 \coloneqq t_{\mathtt{salt},3,e,0}$

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$\mathsf{Expt}^{\mathrm{cavc},b}_{\mathsf{MQOM},\mathcal{A}}(\lambda)$

1 : $(\mathtt{vk},\mathtt{sk}) \leftarrow \mathsf{Gen}(1^\lambda)$
2 : $(\mathtt{vk},x) \coloneqq \mathsf{Parse}(\mathtt{sk})$
3 : $\delta \coloneqq \mathsf{F2B}(x)[0:\lambda]$
4 : $(\mathtt{mseed}_{\mathtt{eq}},\{\hat{y}_i\}) \coloneqq \mathsf{Parse}(\mathtt{vk})$
5 : $(\{\hat{A}_i\},\{\hat{b}_i\}) \coloneqq \mathsf{ExpandEquations}(\mathtt{mseed}_{\mathtt{eq}})$
6 : $(\mathtt{state},\{\mathtt{salt}^{(q)},\mathbf{i}^{*(q)},\}_{q\in[Q]}) \leftarrow \mathcal{A}_0^{E}(\mathtt{vk})$
7 : **assert** no collision in tweaks $t_{\mathtt{salt},k,e,j}$
8 : **for** $q \in [Q]$ **do**
9 : **if** $b=0$ **then**
10 : $\mathsf{out}^{(q)} \coloneqq \mathcal{O}_{\mathrm{real}}(\mathtt{salt}^{(q)},\mathbf{i}^{*(q)})$
11 : **else** $\mathsf{out}^{(q)} \coloneqq \mathcal{O}_{\mathrm{ideal}}(\mathtt{salt}^{(q)},\mathbf{i}^{*(q)})$
12 : $b' \leftarrow \mathcal{A}_1^{E^{\pm}}(\mathtt{state},\{\mathsf{out}^{(q)}\}_{q\in[Q]})$
13 : **return** b'

$\mathcal{O}_{\mathrm{ideal}}(\mathtt{salt},\mathbf{i}^*)$

1 : **for** $e \in [\tau]$ **do**
2 : $i^* \coloneqq \mathbf{i}^*[e]$
3 : $\mathtt{path}[e][1] \leftarrow \{0,1\}^\lambda$
4 : **for** $j=2,\dots,d$ **do**
5 : $b \coloneqq \mathsf{copath}_j(i^*) \bmod 2$
6 : $S_{j-1} \coloneqq \bigoplus_{j'\in[1:j]} \mathtt{path}[e][j']$
7 : $\mathtt{path}[e][j] \leftarrow \{0,1\}^\lambda$
8 : $S_d \coloneqq \bigoplus_{j'\in[1:d+1]} \mathtt{path}[e][j']$
9 : $c_0 \leftarrow \{0,1\}^\lambda$
10 : $c_1 \leftarrow \{0,1\}^\lambda$
11 : $\mathtt{com}[e][i^*] \coloneqq c_0 \,\|\, c_1$
12 : $\mathtt{pdecom}[e] \coloneqq (\mathtt{path}[e],\mathtt{com}[e][i^*])$
13 : $(\mathtt{com}[e],\mathtt{seed}[e],\mathtt{tape}[e]) \coloneqq \mathsf{MQOM.Recon}^{E}(\mathtt{salt},\mathtt{pdecom}[e],e,i^*)$
14 : **for** $e \in [\tau]$ **do**
15 : $\Delta_x^{(1)}[e] \leftarrow \mathbb{F}^{n-\lambda/k}$
16 : $\alpha'_0[e] \leftarrow \mathbb{K}^{\eta'}$
17 : Compute $\Gamma' \coloneqq [\Gamma; O_{(\eta'-\eta)\times\hat{m}}] \in \mathbb{K}^{\eta'\times\hat{m}}$
18 : **return** $(\mathtt{pdecom},\Delta_x^{(1)},\alpha'_0)$

$\mathcal{O}_{\mathrm{real}}(\mathtt{salt},\mathbf{i}^*)$

1 : **for** $e \in [\tau]$ **do**
2 : $i^* \coloneqq \mathbf{i}^*[e]$
3 : $\mathtt{path}[e][1] \leftarrow \{0,1\}^\lambda$
4 : **for** $j=2,\dots,d$ **do**
5 : $b \coloneqq \mathsf{copath}_j(i^*) \bmod 2$
6 : $S_{j-1} \coloneqq \bigoplus_{j'\in[1:j]} \mathtt{path}[e][j']$
7 : $\mathtt{path}[e][j] \coloneqq E(t_{\mathtt{salt},2,e,j-1}, S_{j-1}\oplus\delta) \oplus \sigma(S_{j-1}\oplus\delta) \oplus b\cdot(S_{j-1}\oplus\delta)$
8 : $S_d \coloneqq \bigoplus_{j'\in[1:d+1]} \mathtt{path}[e][j']$
9 : $\mathtt{seed}[e][i^*] \coloneqq S_d \oplus \delta$
10 : $c_0 \coloneqq E(t_{\mathtt{salt},0,e,0}, S_d\oplus\delta)\oplus\sigma(S_d\oplus\delta)$
11 : $c_1 \coloneqq E(t_{\mathtt{salt},1,e,0}, S_d\oplus\delta)\oplus\sigma(S_d\oplus\delta)$
12 : $\mathtt{com}[e][i^*] \coloneqq c_0 \,\|\, c_1$
13 : $\zeta_0 \coloneqq E(t_{\mathtt{salt},3,e,0}, S_d\oplus\delta)\oplus\sigma(S_d\oplus\delta)$
14 : $\zeta_1 \coloneqq E(t_{\mathtt{salt},3,e,1}, S_d\oplus\delta)\oplus\sigma(S_d\oplus\delta)$
15 : $\mathtt{tape}[e][i^*] \coloneqq \zeta_0 \,\|\, \zeta_1$
16 : $\mathtt{pdecom}[e] \coloneqq (\mathtt{path}[e],\mathtt{com}[e][i^*])$
17 : $(\mathtt{com}[e],\mathtt{seed}[e],\mathtt{tape}[e]) \coloneqq \mathsf{MQOM.Recon}^{E}(\mathtt{salt},\mathtt{pdecom}[e],e,i^*)$
18 : **for** $i \in [N]$ **do**
19 : $m \coloneqq \mathtt{seed}[e][i] \,\|\, \mathtt{tape}[e][i] \in \{0,1\}^{3\lambda}$
20 : $\bar{x}[e][i] \coloneqq \mathsf{B2F}(m[0:kn]) \in \mathbb{F}^n$
21 : $\bar{u}'[e][i] \coloneqq \mathsf{B2K}(m[kn:3\lambda]) \in \mathbb{K}^{\eta'}$
22 : $u'_0[e] \coloneqq -\sum_{i\in[N]} \omega_i \cdot \bar{u}'[e][i] \in \mathbb{K}^{\eta'}$
23 : $x_0[e] \coloneqq -\sum_{i\in[N]} \omega_i \cdot \bar{x}[e][i] \in \mathbb{K}^n$
24 : $\Delta_x[e] \coloneqq x - \sum_{i\in[N]} \bar{x}[e][i] \in \mathbb{F}^n$
25 : $\Delta_x^{(1)}[e] \coloneqq \Delta_x[e][\lambda/k:n] \in \mathbb{F}^{n-\lambda/k}$
26 : Compute $\Gamma' \coloneqq [\Gamma; O_{(\eta'-\eta)\times\hat{m}}] \in \mathbb{K}^{\eta'\times\hat{m}}$
27 : **for** $e \in [\tau]$ **do**
28 : **for** $i \in [\hat{m}]$ **do**
29 : $z_{0,i} \coloneqq x_0[e]^\top \cdot \hat{A}_i \cdot x_0[e] \in \mathbb{K}$
30 : $z_0 \coloneqq (z_{0,0},\dots,z_{0,\hat{m}-1}) \in \mathbb{K}^{\hat{m}}$
31 : $\alpha'_0[e] \coloneqq u'_0[e] + \Gamma'\cdot z_0 \in \mathbb{K}^{\eta'}$
32 : **return** $(\mathtt{pdecom},\Delta_x^{(1)},\alpha'_0)$

Fig. 4 Expanded $\mathsf{Expt}^{\mathrm{cavc},b}_{\mathsf{MQOM},\mathcal{A}}(\lambda)$. $\Gamma \coloneqq I_{\eta\times\eta}$ in the 3-round version and $\Gamma \coloneqq \mathsf{XOF}_8(\mathtt{com}_1) \in \mathbb{K}^{\eta\times\hat{m}}$ in the 5-round version, where $\mathtt{com}_1 \coloneqq \mathsf{Hash}_7(h_{\mathtt{com}},\Delta_x^{(1)})$ and $h_{\mathtt{com}}[e] \coloneqq \mathsf{Hash}_6(\mathtt{com}[e])$.

and $t_1 \coloneqq t_{\mathtt{salt},3,e,1}$ for ease of notation.

Recall that $\alpha'_0[e] \coloneqq u'_0[e] + \Gamma'\cdot z_0 \in \mathbb{K}^{\eta'}$, where $x_0[e] \coloneqq -\sum_i \omega_i\cdot\bar{x}[e][i] \in \mathbb{K}^n$, $u'_0[e] \coloneqq -\sum_i \omega_i\cdot\bar{u}'[e][i] \in \mathbb{K}^{\eta'}$, and $z_0 = (z_{0,0},\dots,z_{0,\hat{m}-1})$ with $z_{0,i} \coloneqq \hat{A}_i(x_0[e]) \in \mathbb{K}$. We also recall that $\bar{x}[e][i] \coloneqq \mathsf{B2F}(\mathtt{seed}[e][i] \,\|\, \mathtt{tape}[e][i][0:kn-\lambda]) \in \mathbb{F}^n$, $\bar{u}'[e][i] \coloneqq \mathsf{B2K}(\mathtt{tape}[e][i][kn-\lambda:2\lambda]) \in \mathbb{K}^{\eta'}$, and $x = \sum_{i\in[N]} \bar{x}[e][i] + \Delta_x[e]$, where $\Delta_x[e] = 0^{\lambda/k} \,\|\, \Delta_x^{(1)}[e] \in \mathbb{F}^n$.

For convenience, we define

$$\tilde{x}_1 \coloneqq \textstyle\sum_{i\neq i^*} \bar{x}[e][i] = x - \bar{x}[e][i^*] \in \mathbb{F}^n, \tag{1}$$

$$\begin{aligned}\dot{x} &\coloneqq \tilde{x}_1 - \omega_{i^*}^{-1}\textstyle(\sum_{i\neq i^*}\omega_i\cdot\bar{x}[e][i]) \in \mathbb{K}^n \\ &= x - \omega_{i^*}^{-1}\cdot\omega_{i^*}\bar{x}[e][i^*] - \omega_{i^*}^{-1}\cdot(\textstyle\sum_{i\neq i^*}\omega_i\cdot\bar{x}[e][i]) \\ &= x - \omega_{i^*}^{-1}\cdot x_0[e],\end{aligned} \tag{2}$$

$$\begin{aligned}\tilde{u}'_0 &\coloneqq -\textstyle\sum_{i\neq i^*}\omega_i\cdot\bar{u}'[e][i] \\ &= u'_0[e] + \omega_{i^*}\cdot\bar{u}'[e][i^*] \in \mathbb{K}^{\eta'}.\end{aligned} \tag{3}$$

Note that the values $\tilde{x}_1$, $\dot{x}$, and $\tilde{u}'_0$ are publicly computable. By definition, we have $\Delta_x[e] = x - \sum_{i\in[N]}\bar{x}[e][i] = \tilde{x}_1 - \bar{x}[e][i^*]$.

For notational convenience, define $\delta_\sigma \coloneqq \sigma(\delta)\,\|\,\sigma(\delta)$, $\tilde{\delta}_\sigma \coloneqq \sigma(S_d)\,\|\,\sigma(S_d)$, and $\xi \coloneqq E(t_0,\delta\oplus S_d)\,\|\,E(t_1,\delta\oplus S_d)$. Using these, we have

$$\mathtt{tape}[e][i^*] = \zeta_0\,\|\,\zeta_1 = \delta_\sigma \oplus \tilde{\delta}_\sigma \oplus \xi. \tag{4}$$

From this, we obtain

$$\begin{aligned}\Delta_x^{(1)}[e] &= (x - \textstyle\sum_{i\in[N]}\bar{x}[e][i])[\lambda/k:n] \\ &= \tilde{x}_1[\lambda/k:n] - \bar{x}[e][i^*][\lambda/k:n] \\ &= \tilde{x}_1[\lambda/k:n] - \mathsf{B2F}(\mathtt{tape}[e][i^*][0:kn-\lambda]) \\ &= \tilde{x}_1[\lambda/k:n] - \mathsf{B2F}((\delta_\sigma\oplus\tilde{\delta}_\sigma\oplus\xi)[0:kn-\lambda]),\end{aligned} \tag{5}$$

$$\begin{aligned}\alpha'_0[e] &= u'_0[e] + \Gamma'\cdot\hat{A}(x_0[e]) \\ &= \tilde{u}'_0 - \omega_{i^*}\cdot\bar{u}'[e][i^*] + \Gamma'\cdot\hat{A}\big(\omega_{i^*}\cdot(x-\dot{x})\big) \\ &= \tilde{u}'_0 - \omega_{i^*}\cdot\mathsf{B2K}((\delta_\sigma\oplus\tilde{\delta}_\sigma\oplus\xi)[kn-\lambda:2\lambda]) \\ &\quad + \omega_{i^*}^2\,\Gamma'\cdot\hat{A}(x-\dot{x}),\end{aligned} \tag{6}$$

where we use $\hat{A}(ws) = w^2\cdot\hat{A}(s)$ and eq.(2) implies $x_0[e] =$

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$\omega_{i^*}(x - \dot{x})$.

Rearranging eq.(5) and eq.(6), we express ξ as a function of public and recorded values. Specifically, we define functions F_0 and F_1 with range $\{0,1\}^\lambda$ by

$$\begin{aligned}\xi &= \delta_\sigma \oplus \tilde{\delta}_\sigma \\ &\quad \oplus \begin{pmatrix} \mathsf{F2B}(\tilde{x}_1[\lambda/k : n] - \Delta_x^{(1)}[e]) \\ \quad \| \, \mathsf{K2B}\big(\omega_{i^*}^{-1} \cdot (\tilde{u}'_0 - \alpha'_0[e]) + \omega_{i^*} \Gamma' \cdot \hat{A}(x - \dot{x})\big) \end{pmatrix} \\ &=: F_0(x, i^*, \tilde{x}_1, S_d, \dot{x}, \Gamma', \tilde{u}'_0, \Delta_x^{(1)}, \alpha'_0) \\ &\quad \| \, F_1(x, i^*, \tilde{x}_1, S_d, \dot{x}, \Gamma', \tilde{u}'_0, \Delta_x^{(1)}, \alpha'_0). \end{aligned} \tag{7}$$

Finally, we consider an augmented query-answer list $\mathcal{Q}'_{\mathrm{beta}}$ that stores $(t_0, t_1, S_d, \Delta_x^{(1)}[e], \alpha'_0[e]; i^*, \tilde{x}_1, \dot{x}, \tilde{u}'_0, \Gamma')$. We define the corresponding sets of bad transcripts as follows:

Definition 5.2. *We define the following bad events:*

- $\mathsf{Bad}_{\mathrm{beta},F}$*: There exist* $(t_0, t_1, S_d, *, *; \ldots) \in \mathcal{Q}'_{\mathrm{beta}}$ *and* $(t_b, \delta \oplus S_d, *) \in \mathcal{Q}_E$ *for some* $b \in \{0,1\}$.
- $\mathsf{Bad}_{\mathrm{beta},B,b}$ *for* $b = 0, 1$*: There exist* $(t_0, t_1, S_d, \Delta_x^{(1)}[e], \alpha'_0[e]; i^*, \tilde{x}_1, \dot{x}, \tilde{u}'_0, \Gamma') \in \mathcal{Q}'_{\mathrm{beta}}$ *and* $(t_b, *, \xi_b) \in \mathcal{Q}_E$ *such that* $\xi_b = F_b(x, i^*, \tilde{x}_1, S_d, \dot{x}, \Gamma', \tilde{u}'_0, \Delta_x^{(1)}, \alpha'_0)$.

Bad transcripts: impossible collisions in $E(t, \cdot)$*:* We consider potential collisions in $E(t, \cdot)$ arising from pairs of entries in $\mathcal{Q}_{\mathrm{node}}$, $\mathcal{Q}_{\mathrm{com}}$, and $\mathcal{Q}_{\mathrm{beta}}$, produced by $\mathcal{O}_{\mathrm{ideal}}$ (and similarly by $\mathcal{O}_{\mathrm{real}}$). For example, consider two (possibly distinct) entries $(t, b, s, z), (t', b', s', z') \in \mathcal{Q}_{\mathrm{node}}$ such that $t = t'$ and $\sigma(s \oplus \delta) \oplus b \cdot (s \oplus \delta) \oplus z = \sigma(s' \oplus \delta) \oplus b' \cdot (s' \oplus \delta) \oplus z'$. Such collisions would require two entries with the same tweak t. However, since the game aborts upon any tweak collision, these events cannot occur.

Bounding $\Pr[\Theta_{\mathrm{ideal}} \in \mathsf{Bad}]$*:* We define $\mathsf{Bad} := \mathsf{Bad}_{\mathrm{node},F} \cup \mathsf{Bad}_{\mathrm{node},B} \cup \mathsf{Bad}_{\mathrm{com},F} \cup \mathsf{Bad}_{\mathrm{com},B} \cup \mathsf{Bad}_{\mathrm{beta},F} \cup \mathsf{Bad}_{\mathrm{beta},B,0} \cup \mathsf{Bad}_{\mathrm{beta},B,1}$. We now upper-bound $\Pr[\Theta_{\mathrm{ideal}} \in \mathsf{Bad}]$. To this end, we construct a reduction algorithm $\mathcal{A}_{\mathrm{ow}}$ that simulates the ideal experiment as follows:

(1) Receive $\mathtt{vk}$ from its challenger and compute $\hat{F}$ from $\mathtt{vk}$.

(2) Fix a random tape of $\mathcal{A}$ and run it as a deterministic distinguisher. Simulate the ideal experiment while maintaining $\mathcal{Q}_{\mathrm{node}}$, $\mathcal{Q}_{\mathrm{com}}$, $\mathcal{Q}'_{\mathrm{beta}}$, and $\mathcal{Q}_E$.

(3) If the game does not abort, attempt to find x' such that $\hat{F}(x') = 0$ as follows:

- $\mathsf{Bad}_{\mathrm{node},F}$: For each $t = t_{\mathtt{salt},2,e,j-1}$, find $(t, b, s, z) \in \mathcal{Q}_{\mathrm{node}}$ and $(t, s', *) \in \mathcal{Q}_E$. Let $x_0 := \mathsf{B2F}(s \oplus s') \in \mathbb{F}^{\lambda/k}$. For all $x_1 \in \mathbb{F}^{n-\lambda/k}$, check whether $x' := (x_0, x_1) \in \mathbb{F}^n$ satisfies $\hat{F}(x') = 0$, and output x' if so.
- $\mathsf{Bad}_{\mathrm{node},B}$: For each $t = t_{\mathtt{salt},2,e,j-1}$, find $(t, b, s, z) \in \mathcal{Q}_{\mathrm{node}}$ and $(t, *, y) \in \mathcal{Q}_E$. Let $x_0 := \mathsf{B2F}(\sigma^{-1}(y \oplus z) \oplus s)$ if $b = 0$, and $x_0 := \mathsf{B2F}(\bar{\sigma}^{-1}(y \oplus z) \oplus s)$ otherwise. For all $x_1 \in \mathbb{F}^{n-\lambda/k}$, check whether $x' := (x_0, x_1)$ satisfies $\hat{F}(x') = 0$, and output x' if so.
- $\mathsf{Bad}_{\mathrm{com},F}$: For each $t = t_{\mathtt{salt},i,e,0}$, find $(t, s, z) \in \mathcal{Q}_{\mathrm{com}}$ and $(t, s', *) \in \mathcal{Q}_E$. Let $x_0 := s \oplus s'$. For all $x_1 \in \mathbb{F}^{n-\lambda/k}$, check whether $x' := (x_0, x_1)$ satisfies $\hat{F}(x') = 0$, and output x' if so.
- $\mathsf{Bad}_{\mathrm{com},B}$: For each $t = t_{\mathtt{salt},i,e,0}$, find $(t, s, z) \in \mathcal{Q}_{\mathrm{com}}$ and $(t, *, y) \in \mathcal{Q}_E$. Let $x_0 := \sigma^{-1}(y \oplus z) \oplus s$. For all $x_1 \in \mathbb{F}^{n-\lambda/k}$, check whether $x' := (x_0, x_1)$ satisfies $\hat{F}(x') = 0$, and output x' if so.
- $\mathsf{Bad}_{\mathrm{beta},F}$: For each $t_0 = t_{\mathtt{salt},3,e,0}$ and $t_1 = t_{\mathtt{salt},3,e,1}$, find $(t_0, t_1, s, *, *; \ldots) \in \mathcal{Q}'_{\mathrm{beta}}$ and $(t_b, s', *) \in \mathcal{Q}_E$ for some $b \in \{0,1\}$. Let $x_0 := \mathsf{B2F}(s' \oplus s)$. For all $x_1 \in \mathbb{F}^{n-\lambda/k}$, check whether $x' := (x_0, x_1)$ satisfies $\hat{F}(x') = 0$, and output x' if so.
- $\mathsf{Bad}_{\mathrm{beta},B,b}$ for $b = 0, 1$: For each $t_b = t_{\mathtt{salt},3,e,b}$, find $(t_0, t_1, S_d, \Delta_x^{(1)}[e], \alpha'_0[e]; i^*, \tilde{x}_1, \dot{x}, \tilde{u}'_0, \Gamma') \in \mathcal{Q}'_{\mathrm{beta}}$ and $(t_b, *, \xi_b) \in \mathcal{Q}_E$.
  We then recover x' by solving a system of linear equations. By Lemma 5.1, we obtain
  $$(M_\sigma - M'_{\mathtt{vk},\dot{x},\Gamma',b}) \cdot x = c'_b - \mathsf{B2F}(\xi_b + \sigma(S_d)) \in \mathbb{F}^{\lambda/k}.$$
  We solve this system and output a solution x' such that $\hat{F}(x') = 0$. The running time depends on the rank of the matrix $M_\sigma - M'_{\mathtt{vk},\dot{x},\Gamma',b} \in \mathbb{F}^{\lambda/k \times n}$. Under our conjecture (Definition 5.3) below, this matrix has rank $\lambda/k - O(1)$, yielding running time $2^{kn-\lambda} \cdot \mathsf{poly}(\lambda)$.

Therefore, assuming that our conjecture (Conjecture 5.3) holds, we obtain

$$\Pr[\Theta_{\mathrm{ideal}} \in \mathsf{Bad}] \le \mathsf{Adv}^{\mathrm{ow}}_{\mathsf{F}_{\mathsf{MQOM}},\mathcal{A}_{\mathrm{ow}}}(\lambda).$$

How to solve $\xi_b = F_b(...)$ *for* $\mathsf{Bad}_{\mathrm{beta},B,b}$. It is easy to see, when we fix x_1, $\xi_b = F_b(...)$ yields the system of *quadratic* equations on $x_0 \in \mathbb{F}^{\lambda/k}$ because of $\hat{A}(x - \dot{x})$ in eq.(7). Our key observation is that subtracting $\hat{y}$ with an appropriate factor yields the system of *linear* equations on $x_0 \in \mathbb{F}^{\lambda/k}$.

Lemma 5.1. *Suppose that* $\xi_b = F_b(...)$ *holds, then we can construct a system of linear equations on* x *defined as* $(M_\sigma - M'_{\mathtt{vk},\dot{x},\Gamma',b}) \cdot x = c'_b - \mathsf{B2F}(\xi_b + \sigma(S_d))$.

To show this lemma, we first consider the quadratic function of x, $\omega_{i^*}^{-1} \cdot (\tilde{u}'_0 - \alpha'_0[e]) + \omega_{i^*} \Gamma' \cdot \hat{A}(x - \dot{x})$ inside $\mathsf{K2B}(...)$ in eq.(7). Recall that we have $\hat{y} = \hat{A}(x) + \hat{B} \cdot x$. For each $i \in [\hat{m}]$, we have $(\hat{A}(x - \dot{x}) - \hat{y})[i] = (x - \dot{x})^\top \cdot \hat{A}_i \cdot (x - \dot{x}) - (x^\top \cdot \hat{A}_i \cdot x + \hat{b}_i^\top \cdot x) = (-\dot{x}^\top \hat{A}_i^\top - \dot{x}^\top \hat{A}_i - \hat{b}_i^\top) \cdot x + \dot{x}^\top \hat{A}_i \dot{x} = M_i \cdot x + c_i$, where $M_i := -(\dot{x}^\top \hat{A}_i^\top + \dot{x}^\top \hat{A}_i + \hat{b}_i^\top)$ and $c_i := \dot{x}^\top \hat{A}_i \dot{x}$ are publicly computable. Thus, we can write

$$\begin{aligned}\omega_{i^*}^{-1} \cdot (\tilde{u}'_0 - \alpha'_0[e]) + \omega_{i^*} \Gamma' \cdot \hat{A}(x - \dot{x}) - \omega_{i^*} \Gamma' \cdot \hat{y} \\ = M_{\mathtt{vk},\dot{x},\Gamma'} \cdot x + c, \end{aligned} \tag{8}$$

where $M_{\mathtt{vk},\dot{x},\Gamma'} := \omega_{i^*} \Gamma'[M_0; \ldots; M_{\hat{m}-1}] \in \mathbb{K}^{\eta' \times n}$ and $c := \Gamma' \cdot (c_0, \ldots, c_{\hat{m}-1}) + \omega_{i^*}^{-1} \cdot (\tilde{u}'_0 - \alpha'_0[e]) \in \mathbb{K}^{\eta'}$.

Rewriting and applying $\mathsf{B2F}$ to eq.(7) and letting $\mathsf{K2F} := \phi^{-1}$, we have

$$\begin{aligned}&\mathsf{B2F}(\xi \oplus \delta_\sigma \oplus \tilde{\delta}_\sigma) \\ &= (\tilde{x}_1[\lambda/k : n] - \Delta_x^{(1)}[e]) \, \| \, \mathsf{K2F}(M_{\mathtt{vk},\dot{x},\Gamma'} \cdot x + c) \\ &= (\tilde{x}_1[\lambda/k : n] - \Delta_x^{(1)}[e]) \, \| (\mathsf{K2F}(M_{\mathtt{vk},\dot{x},\Gamma'}) \cdot x + \mathsf{K2F}(c)) \\ &= (M'_{\mathtt{vk},\dot{x},\Gamma',0} \cdot x + c'_0) \, \| (M'_{\mathtt{vk},\dot{x},\Gamma',1} \cdot x + c'_1), \end{aligned} \tag{9}$$

where we define

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$[M'_{\mathsf{vk},\dot{x},\Gamma',0}; M'_{\mathsf{vk},\dot{x},\Gamma',1}] \coloneqq [O_{(n-\lambda/k)\times n}; \mathsf{K2F}(M_{\mathsf{vk},\dot{x},\Gamma'})] \in \mathbb{F}^{2\lambda/k\times n}$

and $c'_0 \,\|\, c'_1 \coloneqq (\tilde{x}_1[\lambda/k : n] - \Delta^{(1)}_x[e]) \,\|\, \mathsf{K2F}(c) \in \mathbb{F}^{2\lambda/k}$ with $M'_{\mathsf{vk},\dot{x},\Gamma',b} \in \mathbb{F}^{\lambda/k\times n}$ and $c'_b \in \mathbb{F}^{\lambda/k}$ for $b = 0, 1$.

We then define a linear map $M_\sigma \colon \mathbb{F}^n \to \mathbb{F}^{\lambda/k}$ as $x \mapsto \mathsf{B2F}(\sigma(\mathsf{F2B}(x)[0 : \lambda]))$. Using this definition, the LHS of eq.(9) is $\mathsf{B2F}(\delta_\sigma) + \mathsf{B2F}(\xi + \tilde{\delta}_\sigma) = (M_\sigma \cdot x + \mathsf{B2F}(\xi_0 \oplus \sigma(S_d))) \,\|\, (M_\sigma \cdot x + \mathsf{B2F}(\xi_1 + \sigma(S_d)))$. Finally, we have two systems of *linear* equations on x as

$$(M_\sigma - M'_{\mathsf{vk},\dot{x},\Gamma',b}) \cdot x = c'_b - \mathsf{B2F}(\xi_b + \sigma(S_d))$$

for $b = 0, 1$. To obtain enough information of x from the system, we need to conjecture that the following holds:

Definition 5.3 (Conjecture)**.** *We conjecture that* $\mathrm{rank}_{\mathbb{F}}(M_\sigma - M'_{\mathsf{vk},\dot{x},\Gamma',b}) = \lambda/k - O(1)$ *for all queries.*

Remark 5.1. *We verify our conjecture experimentally as follows: For each* 12 GF(2) *parameter sets, we generate 100 random verification keys* $(A_i, b_i)_{i\in[m]} \leftarrow (\mathbb{F}^{n\times n} \times \mathbb{F}^n)^m$*. On each key, we generate 100 random* $\dot{x} \leftarrow \mathbb{F}^n$ *(and 100 random* $\Gamma \leftarrow \mathbb{F}^{\mu\eta'\times m}$ *for* 5*-round version for* $\omega_{i^*}\Gamma$*), construct* $M_b \coloneqq M_\sigma - M'_{\mathsf{vk},\dot{x},\Gamma',b}$ *for* $b = 0$ *and* 1*, and compute the rank of* M_b*. We totally compute the matrices* M_b $240,000 = 12 \cdot 100 \cdot 100 \cdot 2$ *times, and all of them are full rank. We put "*$-O(1)$*" for a margin.*

6. EUF-CMA Security of modified MQOM

Using Theorem 5.2, we obtain the following theorem for MQOM', the modified MQOM.

Theorem 6.1 (EUF-NMA to EUF-CMA of MQOM' in the ROM+ICM)**.** *Let* Hash_4 *be a random oracle. Let* $\mathsf{H}_{\mathsf{avc}}(s,t) \coloneqq E(t,s) \oplus \sigma(s)$ *with an ideal cipher* E*. Suppose that the conjecture (Definition 5.3) holds. For a PPT adversary* $\mathcal{A}$ *that queries the oracles* X *at most* Q_X *times, there exist PPT adversaries* $\mathcal{A}_{\mathrm{prg}}$*,* $\mathcal{A}_{\mathrm{ow}}$*, and* $\mathcal{A}_{\mathrm{nma}}$ *such that*

$$\begin{aligned}\mathsf{Adv}^{\text{euf-cma}}_{\mathsf{MQOM}',\mathcal{A}}(\lambda) \le\ & \mathsf{Adv}^{\text{euf-nma}}_{\mathsf{MQOM}',\mathcal{A}_{\mathrm{nma}}}(\lambda) + \mathsf{Adv}^{\mathrm{cr}}_{\mathsf{Hash}_2,\mathcal{A}_{\mathrm{hash}}}(\lambda) \\ & + Q_{\mathsf{Sign}} \cdot \mathsf{Adv}^{\mathrm{prg}}_{\mathsf{PRG}_{\mathsf{seed}},\mathcal{A}_{\mathrm{prg}}}(\lambda) + \mathsf{Adv}^{\mathrm{ow}}_{\mathsf{F}_{\mathsf{MQOM}},\mathcal{A}_{\mathrm{ow}}}(\lambda) \\ & + Q^2_{\mathsf{Sign}}/2^{\lambda-12} + Q_{\mathsf{Sign}}(Q_{\mathsf{Sign}} + Q_{\mathsf{Hash}_4})/2^{\lambda}.\end{aligned}$$

The running times of $\mathcal{A}_{\mathrm{nma}}$*,* $\mathcal{A}_{\mathrm{hash}}$*, and* $\mathcal{A}_{\mathrm{prg}}$ *are that of* $\mathcal{A}$ *plus* $\mathsf{poly}(Q,\lambda)$ *with* $Q = Q_{\mathsf{Sign}} + Q_{\mathsf{Hash}_4}$*. The running time of* $\mathcal{A}_{\mathrm{ow}}$ *is that of* $\mathcal{A}$ *plus* $2^{kn-\lambda} \cdot \mathsf{poly}(Q',\lambda)$ *with* $Q' = Q_E + Q_{\mathsf{Sign}} \cdot \tau(9N/2 - 4)$*.*

For the proof, see the full paper [KX26a].

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Appendix

A.1 MQOM

We review the parameter sets with $|x|$ and $|\Delta_x^{(1)}[e]| + |\alpha_0[e]|$ in Table A·1.

- $|\Delta_x^{(1)}[e]| = n \cdot \log_2(|\mathbb{F}|) - \lambda$
- $|\alpha_0[e]| = \eta \cdot \log_2(|\mathbb{K}|) = \mu\eta \cdot \log_2(|\mathbb{F}|)$
- $\lambda = 128, 192, 256$ (for Cat. I, III, V respectively).

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Table A·1 Parameter Sets of MQOM (v.2.1) [BBFR25] with $|x|$ and $|\Delta_x^{(1)}[e]| + |\alpha_0[e]|$.

Parameter Sets	**MQ Param.**		**Proof System Param.**					**Lengths**	
	$\lvert\mathbb{F}\rvert$	$m = n$	τ	N	μ	η	w	$\lvert x\rvert$	$\lvert\Delta_x^{(1)}[e]\rvert + \lvert\alpha_0[e]\rvert$
MQOM2-L1-gf2-short-3r/5r	2	160	12	2048	16	10 / 8	8	160	192 / 160
MQOM2-L1-gf2-fast-3r/5r	2	160	17	256	8	20 / 16	9	160	192 / 160
MQOM2-L1-gf16-short-3r/5r	16	56	12	2048	4	14 / 8	8	224	320 / 224
MQOM2-L1-gf16-fast-3r/5r	16	56	17	256	2	28 / 16	9	224	320 / 224
MQOM2-L1-gf256-short-3r/5r	256	48	12	2048	2	24 / 8	8	384	640 / 384
MQOM2-L1-gf256-fast-3r/5r	256	48	17	256	1	48 / 16	9	384	640 / 384
MQOM2-L3-gf2-short-3r/5r	2	240	18	2048	16	15 / 12	12	240	288 / 240
MQOM2-L3-gf2-fast-3r/5r	2	240	27	256	8	30 / 24	3	240	288 / 240
MQOM2-L3-gf16-short-3r/5r	16	84	18	2048	4	21 / 12	12	336	480 / 336
MQOM2-L3-gf16-fast-3r/5r	16	84	27	256	2	42 / 24	3	336	480 / 336
MQOM2-L3-gf256-short-3r/5r	256	72	18	2048	2	36 / 12	12	576	960 / 576
MQOM2-L3-gf256-fast-3r/5r	256	72	27	256	1	72 / 24	3	576	960 / 576
MQOM2-L5-gf2-short-3r/5r	2	320	25	2048	16	20 / 16	6	320	384 / 320
MQOM2-L5-gf2-fast-3r/5r	2	320	36	256	8	40 / 32	4	320	384 / 320
MQOM2-L5-gf16-short-3r/5r	16	116	25	2048	4	29 / 16	6	464	672 / 464
MQOM2-L5-gf16-fast-3r/5r	16	116	36	256	2	58 / 32	4	464	672 / 464
MQOM2-L5-gf256-short-3r/5r	256	96	25	2048	2	48 / 16	6	768	1280/ 768
MQOM2-L5-gf256-fast-3r/5r	256	96	36	256	1	96 / 32	4	768	1280/ 768

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